

Bushra Jabeen, MLOps Engineer / Data Scientist

(289) 625-6886 | b.jabeen023@gmail.com | Mississauga, ON | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)

Microsoft Certified: Machine Learning Operations Engineer Associate with experience using Python and SQL to clean data, build forecasting models, and create clear reports and dashboards for real projects in government, education, and healthcare. Developed and tested models for prediction and classification, including time series forecasting and image and text data, and checked results carefully to make sure they were reliable and easy to explain. Built end-to-end workflows with data collection, feature preparation, model training, and simple deployment, using tools like Git, Docker, and cloud platforms to keep work organized and repeatable. Worked closely with different teams to understand their questions, turn them into measurable problems, and deliver tables, charts, and written summaries that supported everyday decisions.

Summary of Qualifications

- Used Python, SQL, Pandas, and Power BI to collect, clean, and visualize data on projects like BeamGroup's open data analytics dashboard
- Developed regression, clustering, and neural network models for forecasting, reaching up to 98% accuracy
- Delivered business results like 50% enrollment growth and 80% better financial recovery through data analysis
- Led cross-functional teams on research and training projects, supporting curriculum design and student performance
- Worked with Agile, Git/GitHub, MLflow, Docker, and dashboards to meet tight deadlines

Technical Skills

Programming & Data:	<i>Python, SQL, Pandas, NumPy, JSON, Shell/Bash</i>
Machine Learning:	<i>Regression, Classification, Clustering, Time Series, NLP, Scikit-learn, TensorFlow, PyTorch</i>
Deep Learning & Gen AI:	<i>Image Classification, Object Detection, Neural Networks</i>
MLOps:	<i>Azure ML, MLflow, Docker, CI/CD Pipelines, Model Lifecycle Management, Model Deployment & Monitoring, Model Serving APIs, Experiment Tracking, Automated Retraining, AI Observability, Drift Detection, Hallucination Monitoring, Model Benchmarking, A/B Testing, Production ML Systems</i>
Gen AI & LLMs:	<i>Prompt Engineering, LLM Evaluation, Fine-Tuning, Retrieval-Augmented Generation (RAG), Vector Databases, Hugging Face Transformers, AI Quality Evaluation, Hallucination Detection, Responsible AI, GenAIOps</i>
Cloud Skills:	<i>Azure AI Services, Azure DevOps, Azure Container Registry, Azure ML Pipelines</i>
Data Visualization:	<i>Power BI, Tableau, Matplotlib, Seaborn, Dash</i>
MLOps & DevOps Tools:	<i>ETL Pipelines, Git/GitHub, MLflow, KNIME, Kubernetes, Ansible, Docker, Agile/JIRA</i>
Statistics:	<i>Probability, Descriptive & Inferential Statistics, Time Series Analysis</i>

Education & Certifications

Microsoft Certified: Machine Learning Operations Engineer Associate (AI-300) **May 2026**
Microsoft | Mississauga, ON

Validated Skills: MLOps infrastructure, ML lifecycle management, GenAIOps, observability, model optimization, deployment, monitoring, and AI system operations

Data Talent Program – Certificate **Jul 2025 – Oct 2025**
M2M Tech Connect | Vancouver, BC

A 16-week industry-aligned program designed to build job-ready data professionals. Key highlights include:

- Built end-to-end data workflows with preprocessing, feature engineering, model training, and deployment using Python, TensorFlow, and Azure ML Studio
- Built and tested ML models with regression and clustering to improve prediction accuracy
- Used MLOps concepts and Git for deployment and documentation in team projects

IBM DevOps and Software Engineer **May 2025**
IBM, Coursera | Mississauga, ON

Junior IT Analyst Program **Nov 2024 – Mar 2025**
NPower Canada | Toronto, ON

IBM Data Science Professional Certificate **Dec 2023**
IBM, Coursera | Mississauga, ON

Master's in Public Health **Sep 2018 - May 2021**
Govt. College University | Faisalabad, Punjab

Work Experience

Data Scientist – Employer-Led Project

Jul 2025 – Nov 2025

BeamGroup | M2M Data Talent Program | Toronto, ON

- Worked with BeamGroup to build a province-wide analytics dashboard for Ontario's Service System Managers, pulling together public datasets on social and labor indicators
- Built forecasting models and dashboards to support capacity planning and resource allocation decisions, similar to supply chain and logistics analytics use cases
- Collected, cleaned, and merged open-government data using Python (Pandas, NumPy), SQL, and Excel to make socioeconomic metrics consistent across municipalities
- Created an interactive HTML map of Ontario schools, adding detailed info and website links right into the analytics dashboard
- Built and tested predictive ML models for unemployment forecasting, optimized regression models to improve forecasting performance
- Uploaded all code, validation scripts, and docs to a shared GitHub repo for easy reuse and teamwork
- Wrote user stories, tested dashboard features, and suggested UI fixes based on feedback during Agile sprints

Data Analyst/Scientist Fellow Internship

Dec 2023 – Mar 2024

Fellowship.AI | United Kingdom

- Cleaned and analyzed large datasets, fixed missing or bad data, and created visualizations to find patterns and support model building
- Tested multiple ML algorithms, hitting 95% accuracy with K-Means clustering and improving to 98% using stable diffusion and ControlNet models
- Wrote a [blog post](#) explaining the project steps, methods, and results for a wider audience

Data Scientist & Data Transformation Projects

Dec 2023 – Present

Freelance/Open-Source Projects | Mississauga, ON

- Built ETL pipelines with SQL and Python to pull, clean, and reshape data from CSV, APIs, and MySQL for modeling in projects like housing price prediction and stock forecasting
- Added validation, outlier detection (Excel, REGEX, Pandas, NumPy), and schema mapping to ensure clean data; tracked experiments with MLflow and deployed via Docker/Flask
- Trained models with Ridge/Logistic Regression, Random Forest, and auto_arima, achieving 86% accuracy on Fashion MNIST image classification and 83% on SpaceX Falcon 9 landing prediction
- Created NLP pipeline with NLTK for IMDB sentiment analysis (83.75% Bernoulli Naive Bayes, 86% Logistic Regression) and end-to-end SQLite/SQL/Seaborn pipeline containerized in Docker
- Built visualizations with Matplotlib, Seaborn, Tableau, and Power BI for time series like Australian consumption trends; all projects on GitHub
- Packaged models and dashboards for deployment on cloud-based environments using Docker, GitHub, and CI/CD-style workflows

Team Lead & Program Coordinator

Mar 2014 – Jul 2021

Federal Institute of Health Sciences | Lahore, Punjab

- Led a multidisciplinary team to execute projects successfully, boosting team morale and productivity through effective communication and guidance
- Designed and delivered curriculum and training sessions, achieving a 90% success rate in key subject areas through effective teaching methodologies
- Analyzed admission data and implemented a targeted campaign strategy, increasing enrollment by 50%, contributing to the department's growth
- Spearheaded financial recovery initiatives, improving department recoveries by 80% through data-driven decisions
- Directed departmental inspections, achieving a 90% compliance with regulatory and operational benchmarks, ensuring operational excellence
- Streamlined the hiring process for staff, ensuring the selection of skilled professionals aligned with organizational goals, contributing to departmental success

Research & Innovation

- Conducted three graduate-level public health research studies involving statistical analysis, data interpretation, and evidence-based recommendations
- Supervised and reviewed eight student research projects, ensuring methodological accuracy and data quality
- Applied quantitative and qualitative research methods to analyze complex datasets and communicate findings to stakeholders